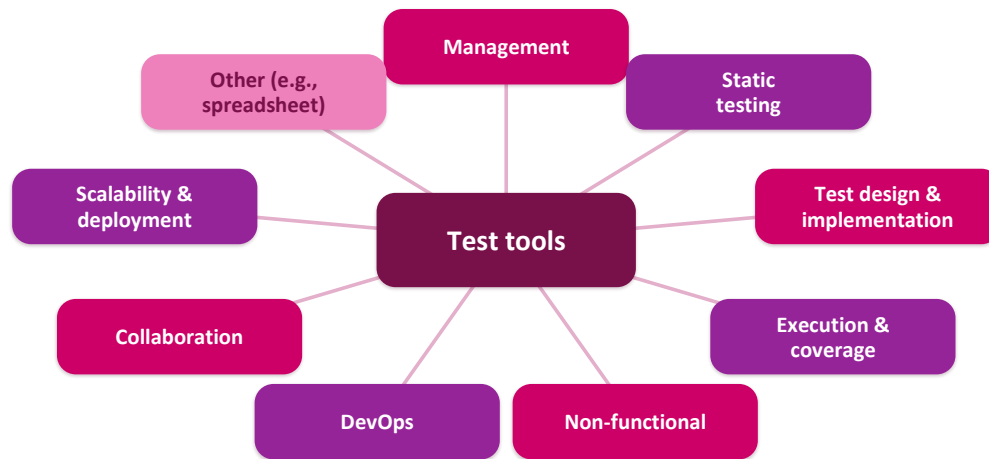


## 6. Test Tools

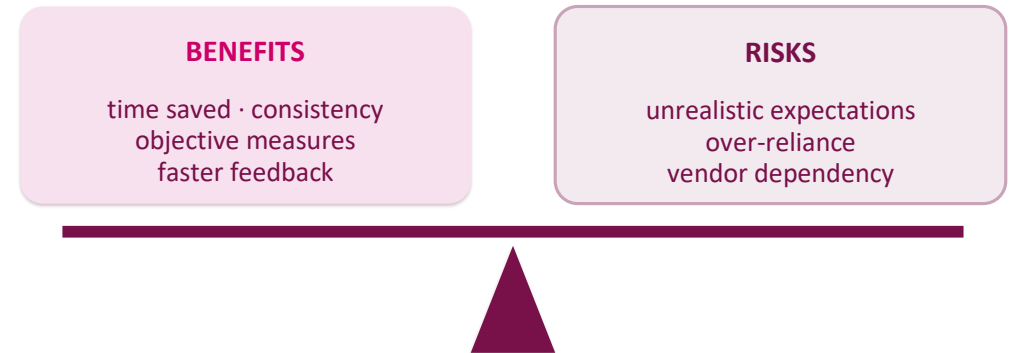
### A. Tool support for testing



Test tools support many different test activities. The main types are:

- **Management tools:** manage the SDLC, requirements, tests, defects and configuration
- **Static testing tools:** support reviews and static analysis
- **Test design and implementation tools:** help create test cases, test data and test procedures
- **Test execution and coverage tools:** run automated tests and measure coverage
- **Non-functional testing tools:** carry out tests that are hard or impossible to do manually
- **DevOps tools:** support the delivery pipeline, builds, CI/CD and workflow tracking
- **Collaboration tools:** help people communicate
- **Scalability and deployment tools:** e.g., virtual machines and containers
- **Any other tool that helps testing:** even a spreadsheet counts

### B. Benefits and risks of test automation



Buying a tool doesn't guarantee success: every tool takes effort to introduce, maintain and train people on.

#### Benefits

- Saves time on repetitive manual work, such as regression testing
- Fewer simple human errors, thanks to consistency and repeatability
- More objective measures, such as coverage
- Easier access to information about testing, for reports
- Faster test execution, giving earlier feedback and a quicker time to market
- More time for testers to design new, deeper tests

#### Risks

- Unrealistic expectations of what the tool can do
- Underestimating the time, cost and effort to introduce and maintain it
- Using a tool where manual testing would be better
- Relying on the tool too much and losing critical thinking
- Depending on a vendor that may go out of business or drop the tool; open-source projects may be abandoned
- The tool may not work with the development platform, or may not meet regulations or safety standards